

**Time:** 5 minutes**Outcome:** Distinguish nominal and real variables and analyze long-run neutrality.**Difficulty:**

Intermediate

Monetary Neutrality



THE CORE IDEA

Monetary neutrality means that a lasting change in the money supply changes nominal variables but not real output or employment in the long run. Nominal variables changed, but real output did not. Monetary neutrality means nominal variables can change without changing real variables in the long run. Separate nominal from real variables. A proportional change in nominal wages and prices can leave purchasing power unchanged.



HOW TO RECOGNIZE IT

- Separate nominal variables, measured in money, from real variables, measured in goods, services, or purchasing power.
- A lasting money-supply change can alter the price level and other nominal values in the long run.
- Under monetary neutrality, long-run real output and employment remain determined by real resources and technology.
- Do not apply long-run neutrality to the short run, when sticky prices can make money affect real activity.

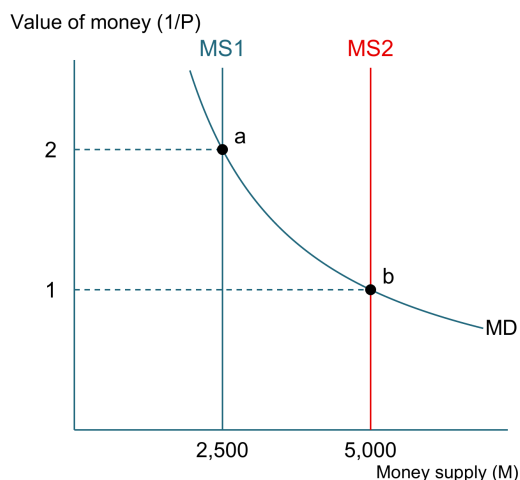


WATCH OUT

Do not assume money changes real variables permanently in the long run. Monetary neutrality means money affects nominal variables in the long run, not real output or employment.



WORKED EXAMPLE: NOMINAL VERSUS REAL EFFECTS



With velocity V and long-run real output Y fixed, money supply doubles from 2,500 to 5,000. The value of money falls from 2 at a to 1 at b , so P rises from $1/2 = 0.50$ to $1/1 = 1.00$: a 100% increase. Prices double while long-run real output does not rise. This is monetary neutrality. Value of money $(1/P)$ is not velocity V in $MV = PY$.



CHECK YOURSELF

A student says, "If money is neutral, monetary policy can never affect real output." What is the best correction?

**READY?**

Return to the game and master this concept.